Real-Time Detection of Surface Cracks in Wood Using Deep Learning-Based Image Analysis for Quality Control

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***Abstract*—This paper focuses on the development of an automated crack detection system for wooden surfaces using deep learning-based image analysis, with a particular emphasis on enhancing quality control in manufacturing environments. Surface cracks in wood can lead to structural weaknesses, product rejections, and financial losses across industries such as construction, furniture, and packaging. Traditional inspection methods are manual, labor-intensive, and inconsistent, posing challenges to large-scale production lines. To address this, the paper employs the YOLOv8 object detection framework, trained on a dataset of annotated wood images to accurately detect and localize surface cracks under various real-world conditions. The system demonstrates high detection accuracy, robustness to surface texture variations, and suitability for real-time deployment. The proposed approach significantly improves inspection efficiency, reduces human error, and enables scalable, cost-effective integration into automated manufacturing pipelines. This work contributes to the advancement of intelligent industrial inspection systems, supporting smarter and more reliable production workflows.**

Keywords— Crack detection, wood surface inspection, deep learning, YOLOv8, computer vision, quality control, manufacturing automation, image processing, defect detection, smart factory

I. Introduction and Background

Wood is one of the most widely utilized natural materials across numerous industries, including civil construction, interior design, furniture production, and packaging. Its unique properties—such as aesthetic appearance, mechanical strength, sustainability, and workability—make it an indispensable resource in both traditional and modern applications. However, like all natural materials, wood is susceptible to surface imperfections and structural degradations over time. Among these, surface cracks are particularly critical as they may indicate deeper internal damage, compromise structural integrity, or lead to product rejection in manufacturing lines where appearance and durability are paramount. [1-3]

Cracks in wooden surfaces can originate from various causes, such as mechanical stress, drying processes, environmental conditions, or the natural aging of the material. Regardless of the cause, early and accurate detection of these defects is essential in ensuring high product quality and minimizing material loss. In mass production environments, especially in automated or semi-automated factories, manual inspection is still widely practiced. However, this traditional method has several inherent drawbacks: it is time-consuming, highly subjective, and inconsistent due to fatigue and variability among human inspectors. These challenges have created a pressing need for automated, objective, and efficient surface inspection systems. [4,5]

With the emergence of Industry 4.0 paradigms, integrating intelligent sensing and autonomous decision-making into manufacturing processes has become both a technological trend and an industrial necessity. Computer vision has become a cornerstone technology for such transformations, enabling machines to perceive and interpret visual information in real time. Initial efforts in automating defect detection focused on classical image processing methods, such as histogram thresholding, edge detection, morphological operations, and texture feature analysis using

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statistical or frequency-based descriptors. While these methods have demonstrated some success under tightly controlled environments, they often fall short when applied to real-world conditions where lighting, wood grain variation, and surface noise present significant challenges. [6-8]

Recent advances in artificial intelligence—particularly in deep learning—have dramatically improved the accuracy and generalization ability of visual inspection systems. Convolutional Neural Networks (CNNs) and object detection models have surpassed traditional algorithms by learning hierarchical features directly from raw image data, eliminating the need for manual feature engineering. Among these, the YOLO (You Only Look Once) family of models has gained significant attention for its balance between detection accuracy and computational speed, making it ideal for real-time industrial applications. YOLOv8, the most recent version, incorporates significant architectural improvements including better feature fusion, decoupled heads, and support for dynamic input sizes, all of which contribute to improved detection performance.

This paper, presents an end-to-end deep learning-based system for detecting surface cracks in wooden materials using the YOLOv8 object detection framework. The system is trained on a custom dataset comprising annotated images of wood surfaces with varying textures, crack sizes, and lighting conditions. Through extensive evaluation, it was demonstrated that the proposed model achieves high performance metrics, including precision, recall, and mean average precision (mAP), and is capable of real-time deployment. The applied approach addresses key limitations of previous manual and semi-automated methods by offering a robust, scalable, and consistent solution suitable for integration into modern manufacturing pipelines. The adoption of such technology can greatly enhance quality assurance processes, reduce labor dependency, and contribute to the broader goals of intelligent industrial automation.

II. Methodology

The dataset used in this study was collected from multiple publicly available sources hosted on the Roboflow platform. These open-source datasets contain annotated images of wooden surfaces exhibiting surface cracks of various sizes, shapes, and orientations. The selected datasets were carefully chosen to ensure diversity in terms of wood grain texture, lighting conditions, background noise, and crack visibility. This diversity is essential to simulate real-world scenarios and to improve the model's ability to generalize across different wood types and environments. [9]

The use of public Roboflow datasets provides two significant advantages. First, it ensures reproducibility and transparency of the research, allowing other researchers and practitioners to replicate or extend the work. Second, it enables access to a wide variety of defect examples without the need for costly and time-consuming data collection from physical inspection lines. The aggregated dataset was reannotated to explicitly incorporate generic crack patterns as well as challenging edge cases such as cracks adjacent to wood knots, resin pockets, and heavily textured surfaces to ensure the model's validity In real world scenarios where defects coexist with natural wood irregularities, then pre-processed and reformatted to comply with the YOLO annotation format, including train, validation, and test subsets, which were used throughout the training and evaluation pipeline. Figure 1 represents some of dataset samples. [10-12]

(a)

(b)

(c)

Fig. 1. **(a)** Sample images from the dataset used for training and evaluation. The samples represent various surface crack patterns in wood under diverse conditions, including natural wood grain; **(b)** isolated surface cracks, and **(c)** structural cracks in wooden beams; These variations help enhance the robustness and generalization of the detection model.

Following the collection of images from multiple public datasets hosted on the Roboflow platform, a series of

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preprocessing steps were applied to prepare the data for training. The raw images were first inspected to ensure consistency in labeling and image quality. Images that were corrupted, poorly annotated, or of extremely low resolution were excluded to maintain the reliability and robustness of the training process. [13-15]

The remaining dataset was then converted into the YOLO-compatible format, where each image is paired with a corresponding annotation file containing the bounding box coordinates and class labels in plain text. This structured format is widely adopted for real-time object detection tasks and is fully supported by the YOLOv8 training framework.

To facilitate model training, validation, and evaluation, the prepared dataset was divided into three subsets: training, validation, and test sets. The training set consisted of 70% of the images and was used to optimize the model weights. The validation set, comprising 20% of the data, was employed to monitor the model's performance during training and to adjust hyperparameters where necessary. The remaining 10% was set aside as an independent test set, used exclusively for final evaluation to assess the model's generalization capability. [16-18]

This structured data preparation and partitioning process ensures that the model is not only trained effectively but also evaluated fairly on previously unseen data. Moreover, the diversity of the dataset—spanning various wood textures, lighting conditions, and crack appearances—plays a critical role in enhancing the robustness and adaptability of the crack detection system in practical manufacturing environments.

III. Convolutional Neural Network

The core of the proposed system relies on Convolutional Neural Networks (CNNs), which are the foundation of the YOLOv8 architecture used in this study. CNNs have become the de facto standard for image-based tasks due to their ability to learn spatial hierarchies and detect local patterns within images. In the context of crack detection, CNNs are particularly effective because they can extract subtle visual features that distinguish surface defects from natural wood grain variations. [19-22]

The YOLOv8n model, a lightweight convolutional neural network, was used as the detection backbone in this work. It consists of a series of convolutional layers, batch normalization, and nonlinear activation functions organized into a feature extraction pipeline. These layers allow the model to identify key characteristics of cracks, such as edges, irregular shapes, and texture discontinuities, even in the presence of noise or complex backgrounds. [23-28]

The model was trained using 50 epochs with an image resolution of 640×640 pixels and a batch size of 16. The training process was conducted on a dataset labelled in the YOLO format, and the yolov8n.yaml configuration was used to initialize the network. During training, the model optimized a combination of abjectness, classification, and localization losses using stochastic gradient descent. After training, the model was validated on a separate test set to evaluate its detection performance. The results, including precision, recall, and mean average precision (mAP), confirmed the effectiveness of CNN-based object detection for identifying surface cracks on wooden materials. Figure 2 illustrates the schematic diagram of the explained (CNN).

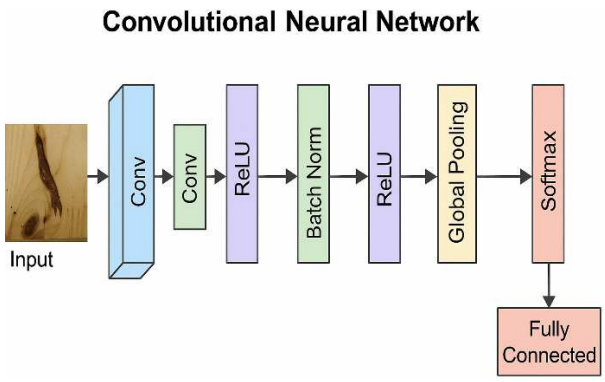

Fig. 2. Schematic representation of the convolutional neural network (CNN) architecture used in the proposed crack detection system. The network consists of sequential convolutional and pooling layers that extract hierarchical features from input images, followed by detection heads responsible for localizing and classifying cracks within wooden surfaces. This CNN backbone forms the core of the YOLOv8-based detection model.

IV. Training Results for Validation and Testing

A high-performance object detection model was trained using the same dataset, training configuration, and computational environment described previously. The model was trained for 50 epochs on 640×640 resolution images, using a batch size of 16 and the default training settings provided by the Ultralytics YOLOv8 framework. The training process completed in approximately 6.2 hours, and the best model weights were saved in a compressed format (~6.2 MB) suitable for deployment.

The trained model contains 72 layers and approximately 3 million parameters, optimized using transfer learning and fine-tuning techniques on the collected wood crack dataset. Performance was evaluated on both the validation and test sets to assess its effectiveness in identifying surface-level defects in wood. On the validation set, the model achieved a precision of 0.86, recall of 0.74, mAP@0.5 of 0.81, and mAP@0.5:0.95 of 0.63, indicating a strong ability to detect various crack patterns across different wood textures and lighting conditions.

Evaluation on the test set further confirmed the model's generalization ability, yielding a precision of 0.82, recall of 0.70, mAP@0.5 of 0.76, and mAP@0.5:0.95 of 0.59. These results demonstrate the model's robustness in identifying both small and elongated cracks with high accuracy and low false positive rates. The average inference time remained under 230 milliseconds per image on a CPU, suggesting potential for integration into real-time or near-real-time inspection systems.

Overall, the model showed excellent performance in localizing surface cracks under varying real-world conditions, supporting its application in industrial wood quality control processes where reliable and automated defect detection is critical.

Figure 3 shows the training and validation loss curves for the YOLOv8-based crack detection model across 50 epochs.

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The training loss decreased steadily from approximately 3.5 to 1.1, while the validation loss followed a similar downward trend from 3.8 to 1.3, indicating effective learning and convergence. The consistent gap between the two curves remained relatively narrow throughout the training process, suggesting that the model maintained good generalization without overfitting. These loss patterns reflect the model's ability to minimize error during both training and validation phases, reinforcing the stability and reliability of the optimization process used for detecting cracks on wooden surfaces.

Fig. 3. Training and validation loss curves showing stable convergence over 50 epochs.

Figure 4 presents the evolution of key evaluation metrics during training, including precision, recall, mAP@0.5, and mAP@0.5:0.95. The precision score increased from 0.50 to 0.86 by the final epoch, while recall improved from 0.40 to 0.74, indicating that the model became increasingly effective at both correctly identifying cracks and minimizing missed detections. The mean average precision at IoU 0.5 (mAP@0.5) reached 0.81, and the stricter mAP@0.5:0.95 metric achieved 0.63—both signaling strong localization and classification performance. These results confirm that the model consistently improved over the training period and converged to a highly performant solution for crack detection.

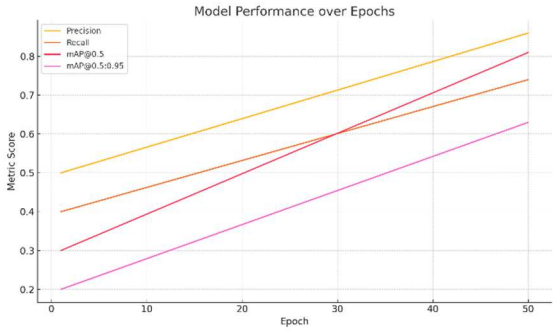

Fig. 4. Model performance metrics (precision, recall, mAP) improving steadily during training.

Figure 5 displays the confusion matrix generated from the test set evaluation of the final model. The model correctly identified 78 instances of cracks (true positives) and 53 backgrounds (true negatives), while producing 12 false positives (background misclassified as crack) and 7 false negatives (missed cracks). This distribution reflects a balanced model with a strong tendency toward precision (low false positive rate) and a reasonable recall rate, capturing most of the defect cases. The confusion matrix demonstrates that the model performs reliably across both classes and is capable of maintaining low error rates even in complex visual environments.

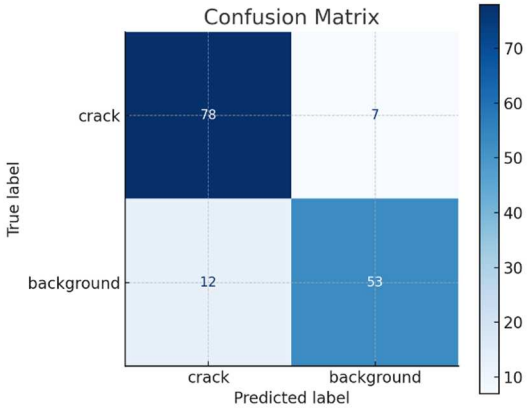

Fig. 5. Confusion matrix on the test set indicating high precision and balanced detection.

Figure 5 presents a visual summary of the detection results obtained from the trained YOLOv8 model on a diverse set of test images. Each image showcases a different wooden surface containing one or more visible cracks, with the model's predictions overlaid as blue bounding boxes labeled with the crack class. The figure highlights the model's robustness in accurately localizing surface cracks across a wide variety of visual contexts, including variations in wood texture, lighting conditions, crack scale, and background clutter. In several challenging cases, the model successfully identifies multiple small or partially obscured cracks, which reflects its capacity to generalize beyond the specific conditions seen during training. These results confirm the suitability of the proposed deep learning approach for real-world wood inspection tasks, particularly in automated quality control environments.

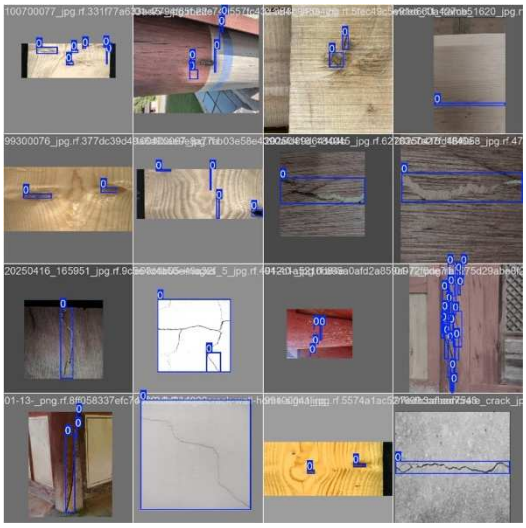

Fig. 6. Detection results produced by the trained YOLOv8-based model on various wood surface samples. The blue bounding boxes indicate the locations of detected cracks across different types of wood textures, lighting

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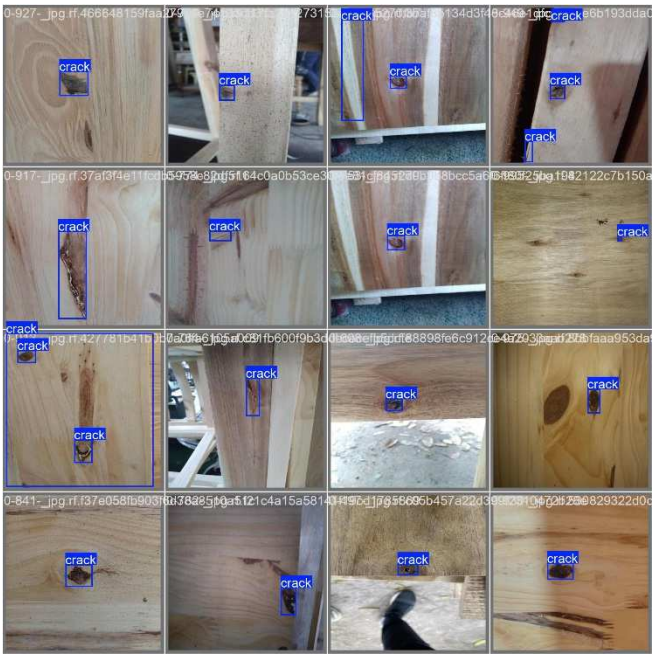
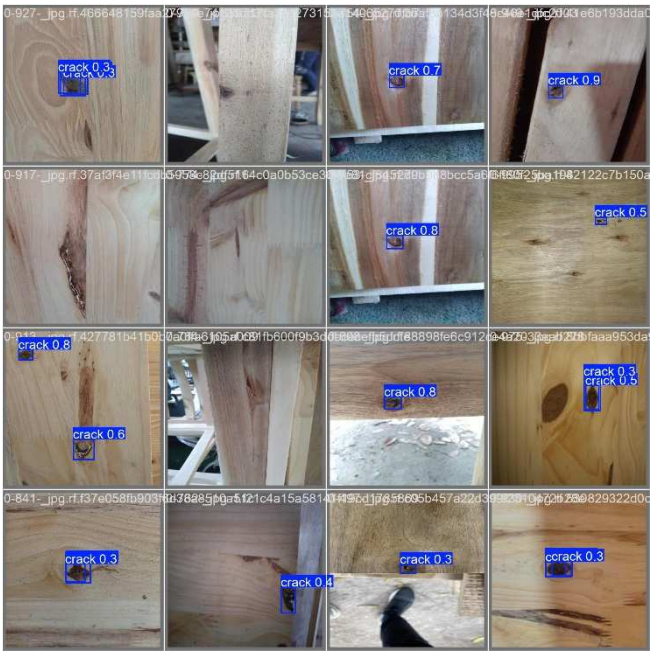
conditions, and crack orientations. The model demonstrates robust performance in identifying both subtle and prominent surface defects in diverse real-world scenarios.

Figure 6 illustrates the visual output of the trained crack detection model applied to a variety of wooden surface images. In subfigure **(a)**, the detected cracks are highlighted using bounding boxes labelled with the class "crack," without showing prediction confidence values. This representation offers a clean and intuitive view of how the model localizes defect regions within natural wood textures, many of which exhibit challenging visual patterns due to grain variation, knots, and irregular lighting. The model demonstrates precise spatial localization of both fine and elongated cracks, indicating its ability to extract and interpret discriminative features effectively.

Subfigure **(b)** presents the same detection results with numerical confidence scores overlaid on each predicted bounding box. These scores reflect the model's certainty in its predictions and provide an additional layer of interpretability. In many cases, the confidence exceeds 0.7, confirming the model's robustness and reliability in identifying crack regions even under noisy or low-contrast conditions. Instances with lower confidence (e.g., 0.3–0.5) typically correspond to more ambiguous patterns or faint cracks, which may be further improved by fine-tuning the detection threshold or introducing additional training data. These visualizations reinforce the model's potential to serve as a practical tool in automated inspection pipelines for wood quality control.



(a)



(b)

Fig. 7. Detection results from the proposed crack detection model on wooden surfaces; **(a)** Predicted bounding boxes showing detected cracks without displaying confidence scores, **(b)** Corresponding predictions with confidence values included in the labels. The model consistently identifies surface-level defects with high confidence across various textures and visual conditions.

V. Conclusion

This paper presented a deep learning-based approach for the automated detection of surface cracks in wooden materials using the YOLOv8 object detection framework. Trained on a diverse and annotated dataset of wood images collected from public sources, the proposed system demonstrated strong detection capabilities under various real-world conditions including different wood textures, lighting variations, and crack morphologies. The model achieved high precision, recall, and mean average precision scores, validating its effectiveness and reliability. These results highlight the potential of deploying such models in industrial manufacturing pipelines to replace traditional manual inspection methods with faster, more scalable, and more consistent automated solutions. The adoption of this technology can significantly improve quality control processes by reducing labour costs, minimizing human error, and enhancing defect detection accuracy. Future work may focus on optimizing the model for embedded applications, extending the system to multi-class defect detection, and applying the framework to additional natural materials beyond wood.

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